

Approaches to derive strain transfer efficiency from LDV Data

The goal is to determine the strain transfer efficiency $\eta = \delta x_l / \delta L$: the fraction of the axial displacement between shaker head and clamp (δL) that is transmitted as axial elongation of the cable (δx_l). I've briefly described the approach that we discussed in the lab (geometric arc-length calculation) as well as another possible approach that is derived from the theory (probably similar to what Claude suggested to you). The theoretical approach is a bit more elegant but you should be cautious because it's only applicable when the shape of the cable deformation corresponds reasonably well to the chosen initial mode shape.

Approach 1: Geometric Arc-Length (linearized)

Idea. The cable's axial elongation δx_l equals the change in its geometric arc-length. This can be computed directly from the 3-component LDV displacements at each measurement point, projected onto the local cable direction, without any assumption about the cable's mode shape.

Implementation. $\mathbf{P}_i^0$ is the initial 3D position of a measurement point i and $\mathbf{u}_i(t)$ is its displacement (obtained by integrating the LDV velocity). The initial length of segment i in the reference configuration is

$$l_i^0 = |\mathbf{P}_{i+1}^0 - \mathbf{P}_i^0|. \quad (1)$$

The change in length of segment i is

$$\delta d_i(t) = |\mathbf{P}_{i+1}^0 + \mathbf{u}_{i+1}(t) - (\mathbf{P}_i^0 + \mathbf{u}_i(t))| - l_i^0 = l_i(t) - l_i^0, \quad (2)$$

and the total axial elongation is

$$\delta x_l(t) = \sum_i \delta d_i(t). \quad (3)$$

The strain transfer efficiency then follows as

$$\eta(t) = \frac{\delta x_l(t)}{\delta L(t)}, \quad (4)$$

where $\delta L(t)$ is the displacement of the shaker head.

Small note on the implementation. The formula above computes a small change δd_i as the difference of two large, nearly equal numbers ($l_i(t)$ and l_i^0). This can introduce numerical noise, because the relative precision of the result is much worse than the relative precision of the inputs. For small displacements (which we are dealing with), a linearized version might be preferable, which avoids this by working directly with the small displacements, never forming the large intermediate lengths. Defining the unit vector along segment i in the reference configuration,

$$\hat{\mathbf{e}}_i = \frac{\mathbf{P}_{i+1}^0 - \mathbf{P}_i^0}{l_i^0}, \quad (5)$$

the linearized segment length change is then

$$\delta d_i(t) \approx \hat{\mathbf{e}}_i \cdot (\mathbf{u}_{i+1}(t) - \mathbf{u}_i(t)). \quad (6)$$

This is the first-order Taylor expansion of the exact formula and is valid when $|\mathbf{u}| \ll l_i^0$, which should be the case (to be checked, but I think even with the relatively large strains in the new setup this should still be valid).

Approach 2: Mode-Shape Projection

Idea. The theoretical model decomposes cable deformation into an axial mode x_l and a flexural mode x_f . From equation (13) of the model, the axial elongation is

$$\delta x_l = \delta L + C_1 x_{f0} \delta x_f, \quad (7)$$

where $C_1 = \int_0^L (\phi')^2 dx$ is a geometric constant of the mode shape $\phi(x)$, x_{f0} is the initial midpoint sag, and δx_f is the change in midpoint deflection. All three quantities (δL , x_{f0} , δx_f) can directly be determined from the measurements under the condition, that the cable shape follows a mode shape (e.g., a symmetric shape with maximum deflection (displacement from a straight centerline) at the midpoint of the hanging segment).

Implementation. Fit the assumed mode shape $\phi(x) = \frac{1}{2}(1 - \cos(2\pi x/L))$ to the initial 3D positions of the measurement points to obtain x_{f0} (the fitted amplitude of the deflection at the midpoint of the hanging cable segment). You should fit this in 3D, so the maximum deflection can be in any direction in the y-z-plane. For each time step, fit the same function to the displaced positions to obtain $x_{f0} + \delta x_f$ (the deflection at the midpoint of the hanging cable segment during the deformation, δx_f is the change in deflection). Then evaluate δx_l from equation (5) and $\eta = \delta x_l / \delta L$.

A least-squares fit of $\phi(x)$ over all measurement points probably helps to remove point-wise LDV noise which could make this approach more robust. However, the result is only valid if the actual cable shape is well described by $\phi(x)$. If the initial cable shape is not well described by a symmetric function, or if the time dependent deformations are not well described by the same symmetric function (as we had already seen a bit in the LDV animations, where we had also seen nodes in the middle of the segment with higher order mode deformation) then this approach will not give you correct estimates of the cable elongation.

If the cable perfectly follows a mode shape, then both approaches are theoretically identical. I would suggest to implement both and comparing the results.